GRADE DISTRIBUTION ANALYSIS USING PYTHON

AIM

To write a Python program to represent the distribution of students across seven grades using a pie chart and a bar chart, identify the grades with the maximum and minimum number of students, and calculate the percentage of students who secured grades A and S combined.

ALGORITHM

1. Start the program.
2. Store the grades F, E, D, C, B, A and S in a list and their corresponding student counts in another list.
3. Calculate the total number of students using sum().
4. Draw a pie chart using the student counts, display grade labels and percentage values, and add a title and legend.
5. Draw a bar chart with grades on the X-axis and number of students on the Y-axis.
6. Display the exact student count above each bar and add a title and legend.
7. Find the maximum and minimum student counts using max() and min(), and identify the corresponding grades.
8. Add the counts for grades A and S, then calculate their combined percentage using (A + S) / total × 100.
9. Display the maximum grade, minimum grade, combined student count for A and S, and the percentage.
10. Stop the program.

PROGRAM

```
import matplotlib.pyplot as plt

# Store grades and number of students
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]
total = sum(students)

# Pie chart
plt.figure(figsize=(7, 6))
plt.pie(students, labels=grades, autopct='%1.1f%%', startangle=90)
plt.title('Percentage Distribution of Students by Grade')
plt.legend(grades, title='Grades', loc='best')
plt.show()

# Bar chart
plt.figure(figsize=(8, 5))
bars = plt.bar(grades, students, color='skyblue', label='Students')
plt.xlabel('Grades')
plt.ylabel('Number of Students')
plt.title('Number of Students in Each Grade')
plt.legend()
```

```
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width() / 2, height,
             str(int(height)), ha='center', va='bottom')
plt.show()

# Maximum and minimum grades
max_students = max(students)
min_students = min(students)
max_grade = grades[students.index(max_students)]
min_grade = grades[students.index(min_students)]

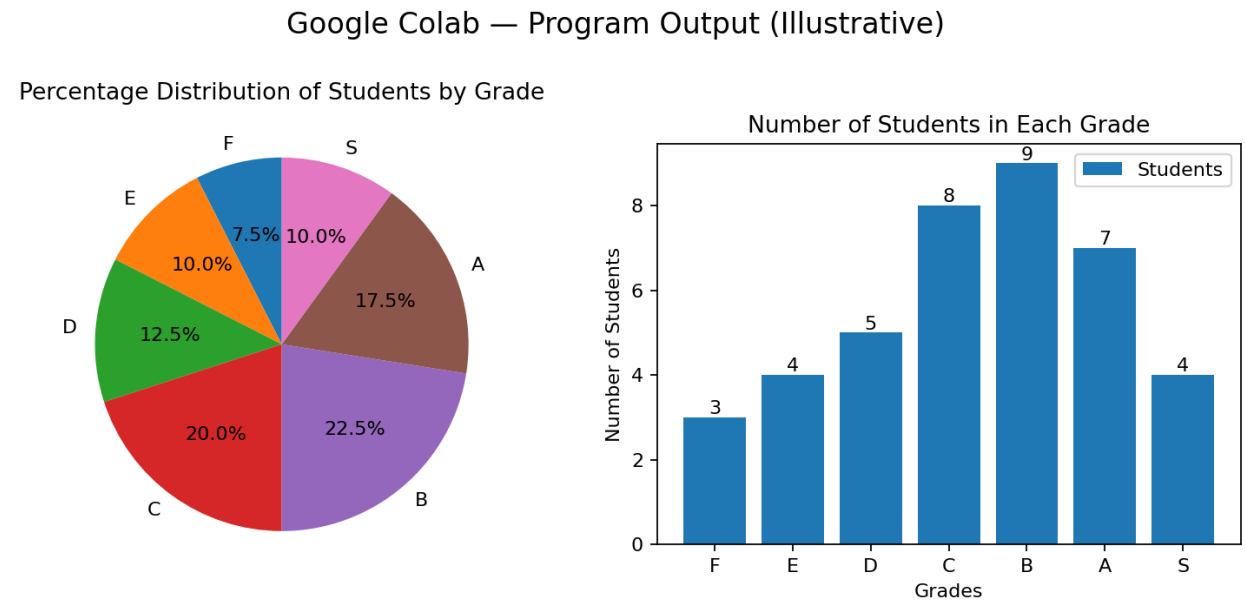
print("Grade with maximum students:", max_grade, "-", max_students)
print("Grade with minimum students:", min_grade, "-", min_students)

# Combined percentage for A and S
a_s_students = students[grades.index('A')] + students[grades.index('S')]
percentage = (a_s_students / total) * 100

print("Students who secured A and S:", a_s_students)
print("Percentage of students with A and S grades: {:.2f}%".format(percentage))
```

PROGRAM OUTPUT SCREENSHOT

The following figure illustrates the charts and printed output produced by running the program in Google Colab.




```
Grade with maximum students: B - 9
Grade with minimum students: F - 3
Students who secured A and S: 11
Percentage of students with A and S grades: 27.50%
```

OUTPUT

Grade with maximum students: B - 9

Grade with minimum students: F - 3

Students who secured A and S: 11

Percentage of students with A and S grades: 27.50%

RESULT

Thus, the Python program was developed to display the percentage distribution of students using a pie chart and compare the number of students in each grade using a bar chart. Grade B has the maximum number of students (9), while grade F has the minimum (3). A total of 11 students secured grades A and S combined, representing 27.50% of the class.